

BFSI BANK — CUSTOMER CHURN ANALYSIS

FINAL BUSINESS RECOMMENDATION — MANAGEMENT REPORT

Prepared for: Head of Retail Banking | Basis: 10,000-customer portfolio, verified across 13 workbook sheets

EXECUTIVE SUMMARY

This bank's retail portfolio of 10,000 customers carries a 20.37% churn rate (2,037 customers) — materially above typical retail banking benchmarks (10-15%) — and the problem is financially worse than the headcount figure suggests: churned customers hold 24.26% of total portfolio Balance (\$185.6 million of \$764.9 million), meaning the bank is disproportionately losing its more valuable relationships, not its most dormant ones.

The analysis identifies five concrete, data-grounded churn drivers (Age, Engagement/Activity, Geography, Gender, Product Count) and rules out three commonly-assumed drivers (CreditScore, Salary, Tenure) that show negligible statistical relationship with churn in this portfolio. A combination of engagement, product-strategy, and geography-specific interventions can realistically cut churn to 11-15% — but hitting a genuinely protective balance-retention target requires an additional, explicitly value-weighted retention track for high-balance customers, since volume-based campaigns alone under-protect the bank's most valuable relationships.

KEY FINDINGS

- Overall churn rate: 20.37% (2,037 of 10,000 customers) — a Tier-1 business problem, not routine attrition.
- Churned customers hold 24.26% of total portfolio Balance, versus 20.37% of headcount — revenue risk exceeds customer-count risk.
- The High Value segment (2,420 customers, Balance>\$100k & Salary>median) holds 41.88% of total Balance and churns at 25.29% — higher than the rest of the portfolio (18.8%). The bank's richest customers are leaving faster than average, not slower.
- The single highest-risk identifiable segment is German Female customers aged 51-60 (129 customers, 73.64% churn rate).
- Engagement is the most directly controllable lever: Inactive members churn at 26.85% vs 14.27% for Active members (1.88x), with no difference in average Balance or Salary between the two groups.
- Zero-balance customers churn LESS (13.86%) than balance-holding customers (24.09%) — churn here is driven by dissatisfied, higher-value customers actively leaving, not dormant low-engagement accounts drifting away.

TOP CHURN DRIVERS (RANKED BY EVIDENCE STRENGTH)

1. Age — strongest numeric correlation with churn (+0.285). Churn rises from 7.56% (<30) to a peak of 56.21% (51-60), then eases to 24.78% (>60).
2. Engagement (Active/Inactive status) — correlation -0.156. The most directly actionable lever; 1.88x churn gap with no confounding Balance/Salary difference.

3. Geography — Germany churns at 32.44% vs 16.15% (France) and 16.67% (Spain), a 16.28-point gap, the largest single categorical spread found in the data.
4. Gender — Female customers churn at 25.07% vs 16.46% for Male, an 8.6-point gap that persists across all three geographies.
5. Product Count — U-shaped, not linear: 2 products is the safe zone (7.58% churn); 1 product is elevated (27.71%); 3-4 products are near-total-loss segments (82.71% / 100%), more consistent with a service failure than normal attrition.

RULED OUT: CreditScore (correlation -0.027), EstimatedSalary (correlation +0.012), and Tenure (churn flat at 17.2%-23.0% across all 10 tenure years) all show negligible relationships with churn in this dataset and should not be used as churn-risk screening variables, despite being intuitive candidates.

HIGH-RISK CUSTOMERS

Two distinct high-risk populations emerge from this analysis, requiring different interventions:

- By DEMOGRAPHIC PROFILE: German Female customers aged 41-60 (combining the 51-60 band at 73.64% churn and the 41-50 band at 52.42% churn) — roughly 459 customers across both bands. This is small enough for high-touch, relationship-manager-level retention outreach.
- By VALUE-AT-RISK: the High Value segment (Balance>\$100k & Salary>median, 2,420 customers, 25.29% churn) — these customers are not necessarily demographically distinct, but their departure causes the largest financial damage per customer lost (\$131,961 average balance per churned High-Value customer, vs \$91,109 portfolio-wide).
- The 3-4 product segment (326 customers, 82.71%-100% churn) is a third, smaller population that looks like a service-quality failure rather than a demographic risk pattern, and warrants root-cause investigation rather than retention marketing.

GEOGRAPHIC FINDINGS

- Germany: 2,509 customers, 32.44% churn (highest), average Balance \$119,730 (highest of the three markets) — the costliest possible combination of high value and high risk.
- France: 5,014 customers (largest market), 16.15% churn (lowest).
- Spain: 2,477 customers, 16.67% churn.
- Ranked by ROI of a uniform 20% country-specific churn reduction: Germany ranks #1 on BOTH customers retained (162.8) and Balance retained (\$19.6M) — confirming it is not just the highest-risk market but the highest-return market to fix.

AGE FINDINGS

- Churn rises sharply through mid-career: <30 (7.56%) -> 30-40 (11.76%) -> 41-50 (33.97%) -> 51-60 (56.21%, peak) -> >60 (24.78%, eases).
- The 41-60 bracket is the bank's highest-risk life stage, likely tied to competing offers, retirement planning, or major life transitions not captured in this dataset.
- Median age of churned customers is 45 vs 36 for retained customers — the gap holds at the individual level, not just in grouped averages.

CREDIT SCORE FINDINGS

- CreditScore shows almost no relationship with churn: Poor band churns at 22.02%, Excellent band at 20.22% — only a 1.8-point spread, and the correlation coefficient (-0.027) is statistically negligible.
- 30.3% of the portfolio (3,034 customers) sits below a 600 CreditScore — a meaningful population for lending/underwriting risk management, but NOT a useful churn-targeting variable.
- Recommendation: exclude CreditScore from any churn-risk model or retention-targeting rule; retain it solely for its intended purpose in credit/lending decisions.

PRODUCT FINDINGS

- Product count shows a U-shaped relationship with churn: 1 product = 27.71% churn (5,084 customers, the largest single product tier), 2 products = 7.58% (lowest, 4,590 customers), 3 products = 82.71% (266 customers), 4 products = 100% (60 customers).
- The 3-4 product segment (326 customers total) looks like a mis-sold or service-failure cohort rather than a cross-sell success story, and should be investigated operationally before any further product-pushing.
- A controlled 1-to-2 product upgrade program for the 5,084 single-product customers is well-supported by the data: moving 40% of them to 2 products would cut overall churn from 20.37% to 16.28% and retain \$40.3 million in Balance.

RECOMMENDATIONS

1. Launch an Inactive-Member Activation Campaign — The single fastest, cheapest, highest-impact lever available. A 30% reduction in Inactive churn alone cuts overall churn from 20.37% to 16.46% and saves \$35.5M in Balance; full activation (Inactive adopting the Active rate) would cut churn to 14.27%. Prioritize the 2,521 Inactive & 1-Product customers first — this sub-segment has the single highest churn rate in the portfolio (36.65%).

2. Build a Dedicated High-Value Customer Retention Track — Not optional — the Integrated Mini-Project proved that volume-based levers alone leave 50%+ of at-risk Balance unprotected. A relationship-manager-led program for the 2,420 High Value customers (Balance>\$100k & Salary>median) is required to meaningfully protect portfolio Balance, not just reduce the headline churn rate.

3. Roll Out a Controlled 1-to-2 Product Upgrade Program — Target the 5,084 single-product customers with a monitored upsell program, capped at 2 products per relationship (the data-proven safe zone). Do NOT extend cross-sell push beyond 2 products — investigate the 3-4 product segment as a service-quality issue instead.

4. Create a Named Germany + Female + Age 41-60 Retention Cohort — This roughly 459-customer cohort combines three independently-significant risk factors into the portfolio's highest-risk segment (up to 73.64% churn). Small enough for high-touch, personalized outreach; the modeled impact of a targeted 50% churn reduction for a comparable segment (Female, 41-50) was 214 additional customers retained and \$18.0M in Balance protected.

5. Investigate Germany as a Market, Not Just a Churn Statistic — Germany ranks #1 by both customers retained and Balance retained under an identical 20% improvement assumption — the clearest signal in this analysis that a geography-specific root-cause investigation (competitor activity, local service quality, pricing) is likely to pay for itself faster than any other initiative.

6. Do Not Build Retention Targeting Around CreditScore, Salary, or Tenure — All three show statistically negligible relationships with churn in this dataset. Resources spent screening or targeting by these variables for retention purposes would be misallocated; reserve CreditScore for its proper use in lending decisions.

EXPECTED BUSINESS IMPACT

Projected outcomes under the three integrated scenarios modeled in this analysis (WhatIf_Integrated sheet), combining Inactive-activation, product cross-sell, and geography-specific levers:

Scenario	New Churn Rate	Balance Protected	Meets 15% Target?	Meets 80% Balance Target?
Moderate Intervention	14.87%	29.0%	YES	NO
Strong Intervention	11.33%	47.7%	YES	NO
Strong + Value-Targeted	6.78%	80.0%	YES	YES

Financial context: at a realistic \$500 cost-per-acquired-replacement-customer and a 2% assumed annual margin on retained Balance, a \$500,000 retention campaign budget breaks even on margin alone by retaining just 344 customers (16.9% of total churned) — a bar the Optimistic Retention scenario alone (~391 customers retained) already clears. Every lever recommended above is therefore not just churn-reducing but economically self-funding under reasonable assumptions.

Bottom line: the 15% churn-rate target is readily achievable with the Strong intervention package alone. Protecting 80% of at-risk Balance — the more financially meaningful and more demanding objective — requires the additional High-Value-targeted track, and should be the primary KPI this program is judged against.

End of Report — BFSI Bank Customer Churn Analysis Project